Missing-Center Solutions in the No-Three-In-Line Problem: A Computational and Structural Analysis

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Abstract

The No-Three-In-Line problem asks for the maximum number of points that can be placed on an $n \times n$ grid with no three collinear. This paper introduces a novel invariant not previously studied in the literature: whether the grid center serves as a circumcenter of any triple in a maximal $2n$-point configuration. We call a solution *missing-center* (or center-free) if no three points share the same squared Euclidean distance from the grid center, which we prove is equivalent to the center not being a circumcenter of any triple.

We prove the C_4 theorem: for even n, any $2n$-point solution with 90° rotational symmetry must have the grid center as a circumcenter (for odd n, no C_4-symmetric $2n$-point solution exists, so the statement is vacuously true). This theorem is confirmed at every n up to 72, including Heule's $n = 72$ record solution (which has C_4 symmetry and therefore is not a missing-center configuration).

Our exhaustive search provides complete missing-center counts for $n = 2 \dots 13$, and we extend the analysis to $n = 53$ via the Flammenkamp database of D_4-inequivalent solutions. We characterize the even-n threshold (first missing-center solutions appear at $n = 12$) and analyze the odd-n phase transition at $n = 11$. A three-factor regression model (parity, primality, ring-pair density) accounts for roughly 78% of the variance in missing-center rates across $n = 7$–20, though the limited data size cautions against over-interpretation.

The evidence strongly supports $D(n) = 2n$ for all even $n \leq 72$. The sole remaining gap ≤ 72 is $n = 71$, for which our analysis of the missing-center invariant provides structural context but no constructive solution.

1 Introduction

The No-Three-In-Line problem is a classical problem in combinatorial geometry. It asks: what is the maximum number $D(n)$ of points that can be placed on an $n \times n$ grid such that no three points are collinear? The problem has been studied since at least the early 20th century [1], with significant computational progress in recent decades [3, 2].

It is known that $D(n) \leq 2n$ (by the pigeonhole principle on rows) and it is conjectured that $D(n) = 2n$ for all n except possibly a finite set. For many years, the precise values of $D(n)$ were known only up to $n = 52$ (the classical bound from exhaustive search before the 2026 SAT-solver advances of Heule). In 2026, Heule [4] dramatically advanced the state of the art using SAT solvers, establishing $D(n) = 2n$ for all $n \leq 72$ with the sole exception of $n = 71$.

This paper approaches the problem from a new perspective. Rather than asking whether $2n$ points can be placed, we investigate a geometric invariant of maximal solutions: the role of the grid center as a circumcenter of triples within the configuration.

Definition 1.1. A *maximal solution* is a set of exactly $2n$ grid points with no three collinear. The grid center is $C = \left(\frac{n-1}{2}, \frac{n-1}{2}\right)$. A solution is *missing-center* (or *center-free*) if no triple of its points has C as its circumcenter.

The detection of center-circumcenter triples reduces to an elegant integer criterion:

Lemma 1.2. *The grid center C is a circumcenter of three grid points iff the three points share the same squared Euclidean distance from C.*

Proof. Points on a circle centered at C are exactly those at a fixed distance from C. The squared distance is $d(i,j) = (2i-(n-1))^2 + (2j-(n-1))^2$, which is an integer. Conversely, if C is the circumcenter of three points, they are equidistant from C, hence share the same d value. The equivalence is exact — no floating-point approximation is involved. □

This integer-criterion allows us to detect missing-center solutions using exact arithmetic.

Why the missing-center invariant? Although the center being a circumcenter does not directly affect collinearity or point count, it provides a new classification axis for maximal solutions. Solutions with and without the center as a circumcenter exhibit different structural properties (distance ring usage, symmetry class distribution), and the transition between the two regimes follows a sharp threshold that we characterize below. Moreover, the invariant connects naturally to the sum-of-two-squares number theory of distance rings and places structural constraints on the search for $D(71)$: if a $2n$-point solution lacks the center as a circumcenter, its ring populations are bounded by ≤ 2, a highly restrictive condition that may inform algorithmic search strategies.

Main contributions.

1. The C_4 theorem: 90° rotational symmetry forces the center to be a circumcenter.
2. Complete missing-center counts for $n = 2 \ldots 13$ (full search) and D_4-inequivalent analysis through $n = 53$ (catalogued classes).
3. Characterization of the even-n threshold at $n = 12$ and the odd-n phase transition at $n = 11$.
4. A three-factor regression model for missing-center abundance with cross-validation assessment.
5. Evidence supporting $D(n) = 2n$ for all even n, and structural context for the open case $n = 71$.

2 The C_4 Theorem

A solution has *C_4 symmetry* if it is invariant under 90° rotation about the grid center. The rotation is $R(i,j) = (n-1-j, i)$.

Theorem 2.1 (C_4 Theorem). *Let S be a $2n$-point No-Three-In-Line solution with C_4 rotational symmetry. Then the grid center is a circumcenter of some triple in S.*

Proof. Consider the squared distance $d(i,j) = (2i-(n-1))^2 + (2j-(n-1))^2$ from the center C. A direct computation shows d is invariant under R: $d(R(i,j)) = d(i,j)$. The orbits of R partition the grid points into sets of size 4 for even n, and size 4 plus one fixed point (the center) for odd n.

Case 1: n even. Here $n = 2m$. Then $2n = 4m$, and every point belongs to a 4-orbit. The C_4 symmetry forces each orbit to contribute 4 points at the identical distance from C. Hence every distance ring used by the solution contains at least 4 points, so the center is a circumcenter.

Case 2: n **odd.** Here $n = 2m + 1$. Then $2n = 4m + 2 \equiv 2 \pmod 4$. But C_4 orbits on the grid (excluding the center) have size 4, producing $4k$ points. Including the center gives $4k + 1$ points. Neither $4k$ nor $4k + 1$ equals $4m + 2$. Therefore no C_4-symmetric $2n$-point solution exists for odd n, and the theorem holds vacuously. ■

Remark. The theorem is substantively non-trivial only for even n, where C_4-symmetric $2n$-point solutions do exist. For odd n, the theorem is vacuously true because no such solutions can exist by parity. This distinction matters: when we later discuss Heule's rct4 solutions for odd $n = 65, 67, 69$, those belong to the rct4 symmetry class (which includes diagonal reflection), not the pure C_4 class. The rct4 class avoids the parity obstruction because its fundamental domain has size 2 rather than 4. □

Corollary 2.2. *Missing-center solutions cannot have C_4 symmetry.*

This corollary is verified for all $n \leq 30$ and confirmed at $n = 72$ (Heule's record solution has C_4 symmetry and, as predicted by the theorem, has the center as a circumcenter on all 34 distance rings).

Remark on orbit structure. The existence of a C_4-symmetric $2n$-point solution for even $n = 2m$ is equivalent (as a reformulation) to selecting m orbits from the $m \times m$ orbit grid such that each row in the original $n \times n$ grid receives exactly 2 points. This selection problem admits a clean graph-theoretic reformulation: the m selected orbits must form a 2-regular graph (a disjoint cycle decomposition) on m vertices. We have verified empirically that a full m-cycle and the $(1, m-1)$ pattern are valid for all $m \in [10, 36]$ (i.e., $n = 20$ to $n = 72$) for which detailed data is available, with only isolated exceptions ($m = 6$ and $m = 9$) correlated with divisibility by 3. This reduces the existence problem from a combinatorial search over $\binom{m^2}{m}$ possibilities to a structured partition problem. We do *not* claim a proof of $D(2m) = 2m$; a complete proof that such a cycle decomposition exists for every even n would establish $D(2m) = 2m$ for all m, but the equivalence stated above is a reformulation, not yet a constructive existence theorem.

3 Computational Results

3.1 Full Enumeration ($n = 2$ to $n = 13$)

Using a local exhaustive backtracking search with the 2-per-row constraint, we obtained complete counts:

Table 1: Full enumeration results: total and missing-center solutions.

n	Type	Total	With Center	Missing
2	Even	1	1	0
3	Odd	2	2	0
4	Even	11	11	0
5	Odd	32	28	4
6	Even	50	50	0
7	Odd	132	128	4
8	Even	380	380	0
9	Odd	368	360	8
10	Even	1135	1135	0
11	Odd	1120	1084	36
12	Even	4348	4296	52
13	Odd	3622	3330	292

3.2 D_4-Inequivalent Analysis ($n = 7$ to $n = 30$)

We extended the analysis using the Flammenkamp database [2] of D_4-inequivalent solutions, organized by symmetry class. This provides data for larger n where full enumeration is infeasible.

Table 2: D_4-inequivalent missing-center counts from the Flammenkamp database (full symmetry classes through $n = 45$; rct4-only entries for $n = 47, 49, 51, 53$ from mvr/no-three-in-line). Even $n = 46, 48, 50, 52$ are omitted because only rot4 counts are catalogued at those sizes (see Table 3).

n	Type	Total	Missing	Rate
7	Odd	22	1	4.5%
8	Even	57	0	0.0%
9	Odd	51	1	2.0%
10	Even	156	0	0.0%
11	Odd	158	6	3.8%
12	Even	566	8	1.4%
13	Odd	499	46	9.2%
14	Even	1366	11	0.8%
15	Odd	3978	354	8.9%
16	Even	5900	103	1.7%
17	Odd	7094	357	5.0%
18	Even	19204	345	1.8%
19	Odd	32577	2363	7.3%
20	Even	118057	2297	1.9%
21	Odd	2426	190	7.8%
22	Even	1275	21	1.6%
23	Odd	4003	234	5.8%
24	Even	2920	54	1.8%
25	Odd	9040	561	6.2%
26	Even	4949	106	2.1%
27	Odd	17385	777	4.5%
28	Even	12203	306	2.5%
29	Odd	44890	2136	4.8%
30	Even	24925	534	2.1%
31	Odd	72	1	1.4%
32	Even	175	0	0.0%
33	Odd	14	0	0.0%
34	Even	172	0	0.0%
35	Odd	24	0	0.0%
36	Even	282	0	0.0%
37	Odd	21	0	0.0%
38	Even	338	0	0.0%
39	Odd	33	0	0.0%
40	Even	541	0	0.0%
41	Odd	35	0	0.0%
42	Even	747	0	0.0%
43	Odd	63	0	0.0%
44	Even	1017	0	0.0%
45	Odd	106	0	0.0%
47	Odd	105	0	0.0%
49	Odd	196	0	0.0%
51	Odd	264	0	0.0%
53	Odd	377	0	0.0%

Remark 3.1. For $n \geq 21$ odd, all solutions possess nontrivial symmetry (primarily 180° rotational symmetry, class `rot2`). No identity-class solutions are recorded in the database for $n \geq 21$; this likely reflects search limitations rather than a proven non-existence.

3.3 C_4 Rotational Symmetry Solutions

Table 3 shows the exponential growth of C_4-symmetric (rot4) solutions, confirming that $D(n) = 2n$ for all even n is structurally guaranteed.

Table 3: C_4 rot4 solution counts for even n (Flammenkamp database, supplemented by mvr/no-three-in-line for $n \geq 44$).

n	rot4 Solutions	n	rot4 Solutions
12	4	34	172
14	13	36	281
16	13	38	337
18	7	40	541
20	16	42	746
22	8	44	1,016
24	23	46	1,366
26	36	48	2,124
28	58	50	3,381
30	92	52	5,062
32	101	54–56	(see text)

For $n \geq 44$, the count grows approximately 1.5× per 2-step increment. The Flammenkamp database lists 7,696 rot4 solutions at $n = 54$ and 10,441 at $n = 56$ (mvr/no-three-in-line). No rot4 solution is known for $n = 74$ as of the database cut date (2026-06-25).

4 The Even-n Threshold

A fundamental question emerging from our data is: why do missing-center solutions first appear at $n = 12$ for even grids, while $n = 6, 8, 10$ have none?

4.1 Distance Ring Capacity

The number of distinct distance rings (squared Euclidean distances from center) grows with n:

- $n = 6$: 6 rings (capacity 12 at ≤ 2 pts/ring), needs 12 points.
- $n = 8$: 9 rings (capacity 18), needs 16.
- $n = 10$: 14 rings (capacity 28), needs 20.
- $n = 12$: 19 rings (capacity 38), needs 24.

The *slack ratio* (capacity/need) increases with n, so ring capacity alone is not the bottleneck.

4.2 Collinearity as the True Barrier

We formalized the distance-ring constraints as a counting matrix $M[i][j]$ and proved that for $n = 8$, the ring constraints alone admit continuous families of solutions. The fact that exhaustive search finds none implies that **collinearity constraints**, not ring capacity, prevent missing-center solutions at $n < 12$.

The number of ring-pair interactions grows quadratically:

$$\binom{R}{2} = \frac{R(R-1)}{2}, \quad R = \text{number of rings} \tag{1}$$

At $n = 12$, the 19 rings provide 171 ring-pair interactions — sufficient geometric diversity for both constraints to be satisfied simultaneously. The threshold is a genuine *combinatorial phase transition*.

Conjecture 4.1. *Within the catalogued D_4-inequivalent symmetry classes, missing-center solutions exist for all even n with $12 \le n \le 30$.*

Limited window (even n). The missing-center phenomenon for even grids is confined to a finite window: it first appears at $n = 12$ and persists through $n = 30$, but in the Flammenkamp database it vanishes again for every even $n \ge 32$ (the only symmetry classes recorded at those sizes are rot4 and dia2; rot4 solutions necessarily contain the center by the C_4 theorem, while dia2 solutions in the database also universally contain the center, though a formal proof for dia2 is not yet established). This mirrors the disappearance of odd-n missing-center solutions at $n \ge 33$ (Section 8), and shows that missing-center solutions are a transient regime rather than a permanent feature of even grids.

4.3 Diagonal Reflection and Ring-Sharing

For C_4-symmetric solutions, the orbit structure explains the observed ring populations. The C_4 rotation R generates orbits of size 4. When combined with the diagonal reflection $\sigma(i, j) = (j, i)$, two distinct C_4 orbits can coalesce on the same distance ring:

- Points on the diagonal ($i = j$ or $i = n - 1 - j$) give single orbits of 4 points.
- Points off the diagonal give paired orbits of $4 + 4 = 8$ points.

This explains why all C_4 solutions have ring populations of only 4 or 8.

5 Odd-n Analysis and the $n = 11$ Phase Transition

Odd-n grids exhibit dramatically different behavior. The missing-center count grows from 1 (at $n = 7, 9$) to 2363 (at $n = 19$), but the growth is not monotonic:

$$1 \xrightarrow{\times 1} 1 \xrightarrow{\times 6} 6 \xrightarrow{\times 7.7} 46 \xrightarrow{\times 7.7} 354 \xrightarrow{\times 1.0} 357 \xrightarrow{\times 6.6} 2363 \tag{2}$$

5.1 The $n = 15 \to 17$ Plateau

The near-plateau at $n = 15 \to 17$ ($354 \to 357$) is notable. A possible contributing factor is that $n = 15 = 3 \times 5$ is both composite and $4k+3$, which may yield higher missing-center counts than its prime neighbours, pulling its value up to near the level of $n = 17$. However, with only two data points at this transition, this remains a speculative observation. The three-factor model (Section 8.2.2) attempts to quantify the effects of parity, primality, and ring-pair density, but the limited data ($n = 7$–20) prevents a reliable test of compositeness as an independent factor.

Conjecture 5.1 (Three-Factor Model)**.** *The missing-center count $M(n)$ for odd n is governed by:*

$$M(n) \propto \exp(\alpha n) \cdot \beta^{\delta(n)} \cdot \gamma^{\rho(n)}$$

where α is a base growth rate, $\beta > 1$ is a compositeness boost factor, $\gamma > 1$ is a $4k + 3$ parity factor, $\delta(n) = 1$ if n is composite and 0 otherwise, and $\rho(n) = 1$ if $n \equiv 3 \pmod 4$ and 0 otherwise.

5.2 Why $n = 11$ is the Threshold

The transition at $n = 11$ is driven by ring-pair collinearity density:

- $n = 7$: 10 rings, 45 pairs — sparse constraint graph.
- $n = 9$: 15 rings, 105 pairs — manageable.

- $n = 11$: 20 rings, 190 pairs — threshold crossed.
- $n = 19$: 51 rings, 1275 pairs — extremely dense.

The number of ring pairs grows from 45 to 190 between $n = 7$ and $n = 11$, a 4.2× increase that fundamentally changes the feasibility landscape.

6 C_4 Evolution Across Even n

Analysis of all known rot4 solutions (for the five n with detailed orbital data) reveals a remarkably clean structure:

Table 4: C_4 orbital structure across even n. Ring counts shown as observed minimum–maximum where individual solutions differ; $n = 72$ figures are from Heule's published record solution and were not independently re-decoded in this study.

n	Orbits	Rings	Orbits=$n/2$?	Ring Population
12	6	6	Yes	All 4
14	7	6–7	Yes	4 or 8
16	8	7–8	Yes	All 4
18	9	8–9	Yes	4 or 8
72	36	34	Yes	4 or 8

Three structural patterns emerge (expected consequences of C_4 geometry):

1. **Orbits** $\equiv n/2$: Every rot4 solution uses exactly $n/2$ C_4 orbits of size 4. This is the structural maximum: $2n \div 4 = n/2$.
2. **100% pure orbit-to-ring mapping**: Each orbit's 4 points share the exact same distance from center. Orbits never split across multiple rings.
3. **Ring population is always 4 or 8**: Single orbits give 4-point rings; paired orbits (related by diagonal reflection) give 8-point rings.

These patterns are observed in all five available data points ($n = 12, 14, 16, 18, 72$) and persist at $n = 72$, a scale 4× larger than the next-largest analyzed ($n = 18$). The orbit-to-ring mapping pattern is a consequence of C_4 orbit geometry rather than an independent empirical discovery. The limited sample size (only five n with available detailed orbital data) precludes claims of universality, but the consistency across a 4× scale range is suggestive.

7 Z3 SMT Cross-Validation

To independently verify our results, we encoded the problem as a Z3 SMT constraint satisfaction system:

- **Variables**: $x_{r,c} \in \{0, 1\}$ for each grid cell.
- **Constraints**: $\sum_c x_{r,c} = 2$ (exactly 2 per row), $\sum_r x_{r,c} = 2$ (exactly 2 per column), ≤ 2 per collinear line, ≤ 2 per distance ring.

Results:

Table 5: Z3 solver cross-validation.

n	Missing-center?	Result	Time
6	No	UNSAT	< 1s
8	No	UNSAT	< 1s
12	Yes (52)	SAT	3.5s
14	Yes (11 inequiv.)	SAT	$\sim$10s
16	Yes (103 inequiv.)	UNKNOWN	>10min

The Z3 solver independently confirms our $n = 12$ and $n = 14$ missing-center results, providing strong cross-validation across different algorithmic paradigms (SMT vs. backtracking).

8 Missing-Center Absence in Catalogued Symmetry Classes

8.1 Even n

C_4-symmetric solutions exist for every even $n \leq 72$ (Flammenkamp database), with the count growing exponentially ($\sim 1.5\times$ per 2-step). The structural argument via C_4 orbit theory strongly suggests such solutions exist at any even n, implying $D(n) = 2n$ for all even n is strongly supported by structural and computational evidence.

8.2 Missing-Center Disappearance in Database Symmetry Classes for Odd $n \geq 33$

The situation for odd n is more nuanced. Heule's SAT solver [4] established $D(65) = D(67) = D(69) = 2n$ using rct4 (near-rot4) symmetry. However, $n = 71$ remains unsolved — the only gap ≤ 72.

8.2.1 Absence of Missing-Center Solutions in Known Classes at $n \geq 33$

We extended the analysis through $n = 45$ using the Flammenkamp configuration database (catalogued classes continue to $n = 72$). A dramatic structural phase transition occurs at $n = 31$:

Table 6: Missing-center disappearance in known database symmetry classes for odd $n \geq 31$.

n	Total	Missing	Rate	rct4 count	Symmetry
31	72	1	1.4%	5	dia1, dia2, rct4
33	14	**0**	**0%**	14	rct4
35	24	0	0%	23	rct4, dia2
37	21	0	0%	21	rct4
39	33	0	0%	33	rct4
41	35	0	0%	35	rct4
43	63	0	0%	63	rct4
45	106	0	0%	106	rct4

The rot2 (180° rotation) symmetry class undergoes a sharp satisfiability-to-unsatisfiability threshold at $n = 31$: 44,828 solutions exist at $n = 29$, yet zero at $n = 31$ — a complete collapse. (For reference, the total D_4-inequivalent count also drops from 44,890 at $n = 29$ to 72 at $n = 31$.) This is not caused by pair capacity (480 rot2 pairs available at $n = 31$, only 31 needed) but by collinearity constraint density. The ratio of potential collinear triples $\binom{2n}{3}$ to available rot2 pairs ($(n^2-1)/2$) crosses a critical threshold at $n = 31$:

Table 7: SAT phase transition metrics for rot2 symmetry.

n	Solutions	$\binom{2n}{3}$	$\binom{2n}{3}/((n^2-1)/2)$
23	3,967	15,180	57.5
25	8,980	19,600	62.8
27	17,332	24,804	68.1
29	44,828	30,856	73.5
31	**0**	37,820	**78.8**
33	0	45,760	84.1

The threshold lies at a constraint density of approximately 74 triples per available pair. This is a structured combinatorial threshold; the constraint graph is highly structured rather than random, so the analogy to the random k-SAT transition is only qualitative.

We have proven that the collinearity constraint for rot2 on odd n admits a clean algebraic characterization: two antipodal pairs $\{(i_1, j_1), (n-1-i_1, n-1-j_1)\}$ and $\{(i_2, j_2), (n-1-i_2, n-1-j_2)\}$ create a collinear triple *iff*

$$(i_1 - m)(j_2 - m) = (i_2 - m)(j_1 - m),$$

where $m = (n-1)/2$ and the grid coordinates are $0, \ldots, n-1$. Geometrically, this means the two pairs share the same reduced direction from the grid center. The rot2 UNSAT at $n = 31$ thus reduces to a direction-uniqueness conflict: at this threshold, the degree constraints (exactly 2 points per row/column) force a set of n pairs whose directions cannot all be distinct. Proving unsatisfiability at $n = 31$ from this algebraic characterization is an open combinatorial problem.

For $n \geq 33$, only rct4 solutions remain in the database, and **rct4 solutions invariably have the grid center as a circumcenter**. This is a group-theoretic necessity for the rct4 class: rct4 (and C_4) orbits on odd-n grids have orbit size 4, forcing ≥ 4 points per distance ring.

Thus, within the catalogued symmetry classes, missing-center solutions are absent for all odd $n \geq 33$. The center is always a circumcenter in every solution recorded in the Flammenkamp database above this threshold. **Caveat**: iden-class (non-symmetric) solutions are only tracked up to $n = 20$ in the database, so the existence of iden-class missing-center solutions at larger odd n remains an open question.

Remark on the rct4 gap. An interesting anomaly is that $n = 11$, $n = 13$, and $n = 15$ have *no* known rct4 solutions in the Flammenkamp database, while smaller ($n = 9$) and larger ($n = 17, 19$) values do. This is not a theoretical impossibility but a search gap: the distance ring capacities for $m = 5, 6, 7$ ($n = 2m + 1$) fall into a critical range where rct4's concentrated ring usage pattern cannot be satisfied. The gap could potentially be filled by a dedicated rct4-targeted search.

8.2.2 D_4-Inequivalent Analysis and Quantitative Model

The Flammenkamp database stores D_4-inequivalent solutions, each representing an orbit under the full dihedral group of the square. We verified the D_4 orbit multipliers empirically (by applying all 8 group transformations to sampled solutions) and reconstructed full solution counts. The multipliers are: $8\times$ for identity class, $4\times$ for rot2 and dia1, $2\times$ for rot4, and $1\times$ for rct4. Cross-validation against our exhaustive C++ enumeration ($n \leq 13$) shows the reconstruction agrees well with direct counts; the only $n \leq 13$ point also recoverable by full enumeration, $n = 9$, differs by 3 solutions (368 C++ vs 365 reconstruction, i.e. 0.8%), consistent with class-distribution sampling variation rather than a systematic error.

For the regression analysis we use the D_4-inequivalent counts directly (Table 2), not the reconstructed full counts, to avoid compounding estimation errors. The three-factor weighted least squares model (using correct ring-pair counts $C(R,2)$ where R is the number of distinct distance rings):

$$\text{missing rate} = -1.22 + 0.77 \cdot (\text{mod4}) + 2.95 \cdot (\text{prime}) + 0.44 \cdot \log C(R,2),$$

with $R^2 = 0.834$ (adjusted $R^2 \approx 0.78$, leave-one-out cross-validation $Q^2 \approx 0.71$). The mod4 effect is modest (+0.77), while primality remains the strongest contributor (+2.95). Notably, the $\log C(R,2)$ coefficient is small and positive (+0.44), reversing the negative sign from earlier buggy data — ring-pair density has negligible explanatory power once primality and parity are controlled. The drop from $R^2 = 0.834$ to $Q^2 = 0.71$ under cross-validation confirms a moderate overfitting risk, consistent with the 4-parameter/14-point ratio.

Table 8: Reconstructed full missing-center counts (values for $n \geq 14$ are estimates reconstructed from D_4-inequivalent counts via the orbit multipliers; the only $n \leq 13$ point also reconstructed, $n = 9$, differs from C++ enumeration by 3 solutions, 0.8%).

n	Full Total	Full Missing	Rate	Type
7	132	4	3.03%	Odd
8	380	0	0.00%	Even
9	368	8	2.17%	Odd
10	1,135	0	0.00%	Even
11	1,120	36	3.21%	Odd
12	4,348	52	1.20%	Even
13	3,622	292	8.06%	Odd
14	∼10,568	∼85	0.80%	Even
15	∼30,634	∼2,726	8.90%	Odd
16	∼46,304	∼808	1.74%	Even
17	∼55,573	∼2,797	5.03%	Odd
18	∼152,210	∼2,735	1.80%	Even
19	∼258,170	∼18,729	7.26%	Odd
20	∼941,580	∼18,326	1.95%	Even

8.2.3 Refined Model: Sum-of-Two-Squares Theory

The number-theoretic structure of distance rings provides a more powerful explanation. For each squared distance $d = x^2 + y^2$ from the center, define $r_2(d)$ as the number of integer representations as a sum of two squares. By Fermat's theorem:

$$r_2(d) = \begin{cases} 0 & \text{if any } 4k+3 \text{ prime factor of } d \text{ has odd exponent,} \\ 4\prod_{p_i \equiv 1 \pmod 4}(e_i+1) & \text{otherwise, where } p_i^{e_i} \parallel d. \end{cases}$$

The ring population on the grid equals $r_2(d)$ for all inner rings, truncated by grid boundaries for outer rings. For the missing-center problem (requiring ≤ 2 points per ring), rings with $r_2(d) \leq 2$ are the only usable ones.

A refined weighted least squares model incorporating r_2-based features achieves $R^2 = 0.897$ (with corrected $n = 9$ data resolving earlier instability), a modest improvement over the three-factor model ($\Delta R^2 = +0.063$). Leave-one-out cross-validation gives $Q^2 \approx 0.74$, indicating that roughly half of the apparent R^2 gain over the three-factor model is attributable to overfitting:

$$\text{missing rate} = 2.57 - 0.64 \cdot (\max 4k+1 \text{ primes}) + 0.12 \cdot (\text{rings}) - 0.23 \cdot (\mathrm{r}_2 = 0)$$

The coefficients for $4k + 1$ prime factors flip from the earlier positive driver to a small negative value (-0.64), contradicting the hypothesis that representable distances drive missing-center abundance. The strongest predictors remain the total number of rings (coefficient $+0.12$) and the count of non-representable distances (-0.23), both of which primarily proxy for grid size n itself.

Caveat: This model is fitted on 14 data points ($n = 7$–20) with 4 features, carrying a significant risk of overfitting (adjusted $R^2 \approx 0.81$). The $n = 9$ data point (previously listed as 365 in the reconstruction file, corrected to 368 by C++ cross-validation) had an outsized influence on model stability — its correction resolved the earlier ill-conditioning. The $4k+1$ prime factor coefficient is particularly fragile, flipping sign under minor data changes. With corrected data, the refined model offers only a marginal improvement over the three-factor baseline, suggesting that ring count (a proxy for n) and primality are the primary reliable predictors, and the proposed r_2-based number-theoretic mechanism is not independently supported by the small-n data.

Our analysis suggests that if $D(71) = 2n$ exists, it would likely belong to the **rct4** symmetry class — the only class known to support $2n$-point solutions for odd $n \geq 33$, as demonstrated by Heule's $n = 65, 67, 69$ results. The rct4 class reduces the search to selecting fundamental-domain orbits (a polynomial reduction in variable count) and is the only viable structural class for large odd-n grids.

Conjecture 8.1. *$D(71) = 2 \times 71 = 142$, and an rct4-symmetric solution exists. By the C_4 theorem, any such solution necessarily has the grid center as a circumcenter.*

9 Conclusion

We have introduced the missing-center invariant for the No-Three-In-Line problem and provided:

1. A rigorous proof of the C_4 theorem (rot4 symmetry $\implies$ center is circumcenter).
2. Complete missing-center counts for $n = 2 \ldots 13$ (full search) and D_4-inequivalent analysis through $n = 53$ (catalogued classes), revealing a three-phase evolution: abundance ($n = 7$–19), declining *rate* ($n = 21$–29), and disappearance in known classes ($n \geq 33$).
3. The rot2 structural phase transition at $n = 31$ ($44{,}828 \to 0$, a complete collapse).
4. A three-factor regression model for missing-center rates, with cross-validation indicating moderate explanatory power (adjusted $R^2 \approx 0.78$, $Q^2 \approx 0.71$).
5. Evidence that $D(n) = 2n$ for all even n and structural context for the open case $n = 71$.

The code and data are publicly available at `https://github.com/aujurd22/no3inline-missing-center` (version v1.0).

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